Generative AI Terminologies: LLM, Agent, and MCP

LLM (Large Language Model):

An **LLM** is an advanced AI system trained to understand and generate human-like text.
It can read, write, and respond to questions in natural language.

Popular LLMs

- ChatGPT
- Google Gemini
- Claude
- DeepSeek

What LLMs Can Do

- Answer questions and explain topics
- Generate code or scripts
- Write emails, blogs, and documents
- Chat naturally with users

Example:

You can ask ChatGPT:

"Write a login test case in Java using Selenium."
It will generate the code for you — but it **won't run the test** itself.

Limitations of LLMs

LLMs are good at *thinking* and *writing*, but they can't *take real actions* in the outside world.

They **cannot**:

- Open a browser or click buttons
- Connect to a database
- Call APIs to get live data

They can only **generate** the instructions or code, not **execute** them.

Agent:

An **Agent** is a system that takes help from an LLM and performs real-world actions using external tools.

Think of it as an **assistant** that actually *does* what the LLM *says*.

Example:
If the LLM says "Run this Playwright test,"
the **Agent** will execute that test in a real browser.

So:

- The **LLM** thinks and plans.
- The **Agent** executes and performs.

MCP (Model Context Protocol):

MCP is a framework that connects LLMs to real-world tools like browsers, databases, and APIs.
It acts as a **bridge** that allows LLMs to send commands to tools and get back the results.

What MCP Can Do

1. **Browser Automation**
    - Open a website, fill forms, and click buttons
    - Example: Selenium MCP or Playwright MCP
2. **Database Operations**
    - Connect to MySQL or PostgreSQL and run queries
    - Example: MySQL MCP or PostgreSQL MCP
3. **API Requests**
    - Send or receive real-time API responses
    - Example: API MCP for calling web services

So, MCP helps the LLM and Agent work together to perform live, practical tasks.

Playwright MCP:

Playwright MCP connects LLMs to browsers using the **Playwright** automation tool (similar to Selenium).

Key Features

- Allows LLMs to control browsers (open sites, fill forms, click elements)
- Understands the web page structure using the **Accessibility Tree** (not images)
- Supports browser commands such as:
    - click()
    - fill('user@example.com')

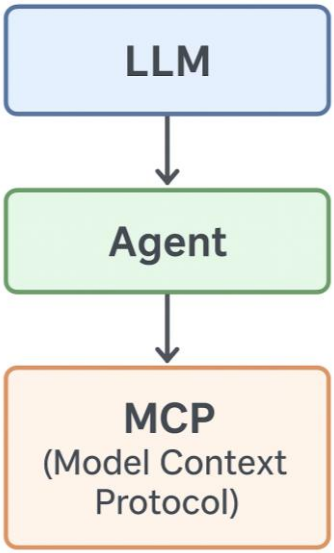
Example

You tell ChatGPT:

"Search for 'Selenium vs Playwright' and click the first result."

Behind the scenes:

- ChatGPT (LLM) sends this instruction to **Playwright MCP**.
- Playwright MCP opens a browser and performs the task automatically.



Summary

Term	In Simple Words	Example Scenario
LLM	The brain that understands and generates text.	GPT-5 explaining "What is API Testing?"
Agent	The assistant that uses the brain to perform actions.	Agent running a Playwright test based on ChatGPT's instructions.
MCP	The bridge that connects the brain and assistant to external tools.	ChatGPT using MCP to open files or run code in PyCharm.